Lesson 32: Convergence Rates, Two-Timescale SA, and RL Applications

AI/ML Mathematics

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1 Motivation and Overview

Lesson 31 established *that* stochastic approximation algorithms converge. This lesson addresses three natural follow-up questions: (1) *how fast* do they converge? (2) what happens when two coupled SA iterations run at *different rates*? and (3) how does the general theory map to specific RL algorithms?

The convergence rate analysis yields a central limit theorem (CLT) for SA and the Polyak–Ruppert averaging trick, which achieves the optimal rate without knowledge of the problem parameters. Two-timescale SA, developed by Borkar [1997], provides the convergence framework for actor-critic algorithms. The linear SA theory gives the sharpest results for TD methods with function approximation.

Why This Lesson Matters for RL

- The **CLT for SA** characterizes the asymptotic variance of RL estimators, explaining why REINFORCE has high variance and motivating baselines.
- **Polyak–Ruppert averaging** underlies target networks (DQN), exponential moving averages (DDPG/TD3), and iterate averaging in policy optimization.
- **Two-timescale SA** is the convergence engine for actor-critic methods: the critic (fast) estimates the value function while the actor (slow) improves the policy.
- **Linear SA** provides the convergence guarantee for TD(0) with linear function approximation and explains the *deadly triad* (divergence under off-policy + FA + bootstrapping).

Prerequisites. Lesson 31 (Robbins–Monro iteration, ODE method, Borkar–Meyn theorem). The martingale CLT, basic matrix analysis (Hurwitz matrices, Lyapunov equations), and familiarity with RL update rules (TD(0), Q-learning, REINFORCE, actor-critic).

2 References

- Borkar [2008], Chapters 4–6: convergence rates, two-timescale SA, and linear SA.
- Borkar [1997]: the original two-timescale SA paper.
- Polyak and Juditsky [1992]: Polyak–Ruppert averaging with optimality proof.
- Kushner and Yin [2003], Chapters 6–10: rate results and averaging, including Markovian noise extensions.
- Borkar and Meyn [2000]: the Borkar–Meyn ODE method applied to RL algorithms.
- Benveniste et al. [2012], Chapter 3: CLT and asymptotic normality for SA.

3 Convergence Rate Analysis

Having established convergence (Lesson 31, Theorem 3.6), we now ask: *how fast* does $x_n \to x^*$?

3.1 CLT Assumptions

Definition 3.1 (CLT Assumptions for SA)**.** Assume the ODE method assumptions (B1)–(B5) from Lesson 31 hold, and additionally:

(D1) h is differentiable at x^* with Jacobian $A = Dh(x^*) \in \mathbb{R}^{d\times d}$.

(D2) A is Hurwitz: all eigenvalues of A have strictly negative real parts.

(D3) The step size is $a_n = a/n$ for some constant $a > 0$ such that $aA + \frac{1}{2}I$ is Hurwitz (i.e., all eigenvalues of aA have real part less than $-1/2$).

(D4) The conditional covariance converges: $\mathbb{E}[M_{n+1}M_{n+1}^T \mid \mathcal{F}_n] \to \Sigma$ in probability as $x_n \to x^*$, for some positive definite Σ.

Remark 3.2. Condition (D3) requires that the step-size constant a be "large enough" relative to the spectral radius of A. If A has eigenvalues $\{\lambda_i\}$, the condition becomes $\mathrm{Re}(a\lambda_i) < -1/2$ for all i. When $d = 1$ and $A = -\lambda$ ($\lambda > 0$), this gives $a > 1/(2\lambda)$.

3.2 The CLT for Stochastic Approximation

Theorem 3.3 (CLT for Stochastic Approximation)**.** *Under the assumptions of Definition 3.1,*

$$\sqrt{n}(x_n - x^*) \xrightarrow{d} \mathcal{N}(0, V),$$

where V is the unique solution of the Lyapunov equation

$$(aA + \tfrac{1}{2}I)V + V(aA + \tfrac{1}{2}I)^T + a^2\Sigma = 0. \tag{1}$$

Proof. Define $e_n = x_n - x^*$ and $u_n = \sqrt{n}\, e_n$. Near x^*, $h(x_n) = Ae_n + o(\|e_n\|)$. With $a_n = a/n$ and $\sqrt{(n+1)/n} = 1 + 1/(2n) + O(1/n^2)$, substituting $e_n = u_n/\sqrt{n}$:

$$u_{n+1} = u_n + \frac{1}{n}\Big(aA + \frac{1}{2}I\Big)u_n + \frac{a}{\sqrt{n}}M_{n+1} + o(1/n).$$

This is a linear stochastic recursion with drift matrix $aA + \frac{1}{2}I$ (Hurwitz by (D3)) and noise covariance $a^2\Sigma$. The covariance $V_n = \mathbb{E}[u_n u_n^T]$ satisfies $V_{n+1} \approx V_n + \frac{1}{n}[(aA + \frac{1}{2}I)V_n + V_n(aA + \frac{1}{2}I)^T + a^2\Sigma]$. At the fixed point, the Lyapunov equation (1) determines the unique $V \succ 0$. The martingale CLT applied to the linearized recursion gives $u_n \xrightarrow{d} \mathcal{N}(0, V)$. □

Remark 3.4 (Optimal Step Size). When $d = 1$ and $A = -\lambda$ ($\lambda > 0$), the Lyapunov equation gives $V = a^2\sigma^2/(2a\lambda - 1)$. Minimizing over $a > 1/(2\lambda)$ yields $a_{\text{opt}} = 1/\lambda$ and $V_{\text{opt}} = \sigma^2/\lambda^2$. In higher dimensions, the optimal a depends on the unknown Jacobian A.

3.3 Polyak–Ruppert Averaging

The optimal step size depends on the Jacobian A at x^*, which is typically unknown. Polyak and Juditsky [1992] proposed averaging the iterates to achieve the optimal rate *without* knowing A.

Theorem 3.5 (Polyak–Ruppert Averaging)**.** *Let $\{x_n\}$ be the SA iterates with step size $a_n = a/n^\alpha$ for $\alpha \in (1/2, 1)$, and define*

$$\bar{x}_n = \frac{1}{n}\sum_{k=1}^{n} x_k.$$

Under assumptions (B1)–(B5) and (D1)–(D4) (with the modified step size), we have

$$\sqrt{n}(\bar{x}_n - x^*) \xrightarrow{d} \mathcal{N}(0,\, A^{-1}\Sigma A^{-T}).$$

This is the optimal asymptotic covariance (matching the Cramer–Rao bound for the root-finding problem).

Proof. Since $x_n \to x^*$ and $h(x^*) = 0$, write $x_{n+1} - x_n = a_n[A(x_n - x^*) + r_n + M_{n+1}]$ where $r_n = o(\|x_n - x^*\|)$. Summing from $k = 1$ to n:

$$x_{n+1} - x_1 = A\sum_{k=1}^{n} a_k(x_k - x^*) + \sum_{k=1}^{n} a_k r_k + \sum_{k=1}^{n} a_k M_{k+1}.$$

An Abel summation argument (using $a_k^{-1} \sim (k/a)^\alpha$ grows polynomially while $a_k(x_k - x^*) \to 0$) shows

$$\sqrt{n}(\bar{x}_n - x^*) = -A^{-1} \cdot \frac{1}{\sqrt{n}} \sum_{k=1}^{n} M_{k+1} + o_P(1).$$

The remainders vanish: $x_{n+1} - x_1 = o(\sqrt{n})$ and $\sum a_k r_k$ is negligible since $\alpha < 1$. By the martingale CLT, $\frac{1}{\sqrt{n}} \sum M_{k+1} \xrightarrow{d} \mathcal{N}(0, \Sigma)$, so $\sqrt{n}(\bar{x}_n - x^*) \xrightarrow{d} \mathcal{N}(0, A^{-1}\Sigma A^{-T})$. □

Polyak–Ruppert Averaging

By averaging the iterates of a slowly decaying SA scheme ($a_n = O(n^{-\alpha})$, $\alpha \in (1/2, 1)$), the averaged iterate $\bar{x}_n$ achieves the optimal convergence rate $O(1/\sqrt{n})$ with the minimal asymptotic covariance $A^{-1}\Sigma A^{-T}$, *without* knowledge of the Jacobian $A = Dh(x^*)$.

4 Two-Timescale Stochastic Approximation

4.1 Motivation

Many RL algorithms involve two coupled iterative updates running simultaneously at different rates. The canonical example is actor-critic: the critic updates the value function (fast timescale) while the actor updates the policy (slow timescale).

4.2 Formulation

Definition 4.1 (Two-Timescale SA)**.** Consider the coupled iterations

$$x_{n+1} = x_n + a_n[f(x_n, y_n) + M_{n+1}^{(1)}], \tag{2}$$

$$y_{n+1} = y_n + b_n[g(x_n, y_n) + M_{n+1}^{(2)}], \tag{3}$$

where $x_n \in \mathbb{R}^{d_1}$, $y_n \in \mathbb{R}^{d_2}$, and the step-size sequences satisfy:

(i) $\sum a_n = \sum b_n = \infty$ and $\sum a_n^2 < \infty$, $\sum b_n^2 < \infty$.

(ii) $a_n/b_n \to 0$ as $n \to \infty$ (*timescale separation*).

The typical choice is $a_n = 1/n$ and $b_n = 1/n^\beta$ with $\beta \in (1/2, 1)$.

Remark 4.2 (Timescale Separation). The condition $a_n/b_n \to 0$ means y uses *larger* step sizes than x, so y moves faster. From the y-perspective, x changes so slowly it appears quasi-static. From the x-perspective, y equilibrates so quickly it appears to have converged.

4.3 Two-Timescale Assumptions

Definition 4.3 (Two-Timescale Assumptions)**.** (E1) **Step sizes:** As in Definition 4.1.

(E2) **Lipschitz drifts:** $f : \mathbb{R}^{d_1} \times \mathbb{R}^{d_2} \to \mathbb{R}^{d_1}$ and $g : \mathbb{R}^{d_1} \times \mathbb{R}^{d_2} \to \mathbb{R}^{d_2}$ are globally Lipschitz.

(E3) **Martingale difference noise:** $\mathbb{E}[M_{n+1}^{(i)} \mid \mathcal{F}_n] = 0$ and $\mathbb{E}[\|M_{n+1}^{(i)}\|^2 \mid \mathcal{F}_n] \leq K(1 + \|x_n\|^2 + \|y_n\|^2)$ for $i = 1, 2$.

(E4) **Stability:** $\sup_n(\|x_n\| + \|y_n\|) < \infty$ a.s.

(E5) **Fast timescale equilibrium:** For each fixed $x \in \mathbb{R}^{d_1}$, the ODE $\dot{y} = g(x, y)$ has a globally asymptotically stable equilibrium $\lambda(x)$, and $\lambda : \mathbb{R}^{d_1} \to \mathbb{R}^{d_2}$ is Lipschitz.

(E6) **Slow timescale stability:** The ODE $\dot{x} = f(x, \lambda(x))$ has a globally asymptotically stable equilibrium x^*.

4.4 Convergence Theorem

Theorem 4.4 (Two-Timescale Convergence (Borkar)). *Under assumptions (E1)–(E6), the two-timescale SA iterates satisfy*

$$x_n \xrightarrow{a.s.} x^*, \quad y_n \xrightarrow{a.s.} \lambda(x^*) =: y^*.$$

Proof. **Phase 1 (fast timescale).** Since $a_n/b_n \to 0$, on any window of fast-timescale length T the total change in x satisfies $\|x_{m_n} - x_n\| \leq C \sum_{k=n}^{m_n-1} a_k \leq CT \sup_{k \geq n}(a_k/b_k) \to 0$. So x is quasi-static on the fast timescale. The y-iteration behaves like SA with frozen x, converging to $\lambda(x)$ by the ODE method. Thus y_n tracks $\lambda(x_n)$: for any $\varepsilon > 0$, $\|y_n - \lambda(x_n)\| < \varepsilon$ for all large n a.s.

Phase 2 (slow timescale). Since $y_n \approx \lambda(x_n)$, the Lipschitz property of f gives $f(x_n, y_n) = f(x_n, \lambda(x_n)) + o(1)$. The x-iteration behaves like SA with drift $f(x, \lambda(x))$, whose ODE has x^* as globally stable equilibrium by (E6). The ODE method gives $x_n \xrightarrow{\text{a.s.}} x^*$, and continuity of λ yields $y_n \to \lambda(x^*) = y^*$. □

Two-Timescale Convergence

In a two-timescale SA system with $a_n/b_n \to 0$:

1. The fast iterate y_n tracks the equilibrium map $\lambda(x_n)$ where $g(x, \lambda(x)) = 0$.
2. The slow iterate x_n converges as if y_n were always at equilibrium, driven by $\dot{x} = f(x, \lambda(x))$.
3. The overall system converges to $(x^*, y^*) = (x^*, \lambda(x^*))$.

5 Linear Stochastic Approximation

The most important special case for RL is *linear* SA.

Definition 5.1 (Linear SA). A *linear stochastic approximation* iteration is

$$x_{n+1} = x_n + a_n[Ax_n + b + M_{n+1}], \tag{4}$$

where $A \in \mathbb{R}^{d \times d}$, $b \in \mathbb{R}^d$, and $h(x) = Ax + b$. The associated ODE is $\dot{x} = Ax + b$ with equilibrium $x^* = -A^{-1}b$ (when A is invertible).

Theorem 5.2 (Convergence of Linear SA). *Suppose A is Hurwitz, the step sizes satisfy the Robbins–Monro conditions, the noise satisfies (B3) from Lesson 31, and the iterates are bounded a.s. Then $x_n \xrightarrow{a.s.} x^* = -A^{-1}b$.*

Proof. Since A is Hurwitz, there exists $P \succ 0$ with $A^T P + PA = -I$. The Lyapunov function $V(x) = (x - x^*)^T P(x - x^*)$ satisfies $\dot{V} = (x - x^*)^T(A^T P + PA)(x - x^*) = -\|x - x^*\|^2$ along ODE trajectories (using $Ax + b = A(x - x^*)$). So x^* is globally asymptotically stable, and the ODE method gives $x_n \xrightarrow{\text{a.s.}} x^*$. □

Corollary 5.3 (Automatic Stability for Linear SA)**.** *For linear SA (4) with A Hurwitz, the boundedness assumption* $\sup_n \|x_n\| < \infty$ *a.s. holds automatically.*

Proof. Apply the Borkar–Meyn theorem (Lesson 31, Theorem 5.2). For $h(x) = Ax + b$, the scaled limit is $h_\infty(x) = \lim_{r\to\infty}(A(rx) + b)/r = Ax$. Since A is Hurwitz, the ODE $\dot{x} = Ax$ has the origin as globally asymptotically stable equilibrium, so (C2) from Lesson 31 is satisfied. □

Theorem 5.4 (Complete Convergence for Linear SA)**.** *For the linear SA iteration (4) with A Hurwitz, Robbins–Monro step sizes, and noise satisfying (B3) from Lesson 31, we have $x_n \xrightarrow{a.s.} -A^{-1}b$ without any additional stability assumptions.*

Proof. Combine Corollary 5.3 (boundedness) with Theorem 5.2 (convergence given boundedness). □

Linear SA: The Hurwitz Condition

For linear SA with drift $h(x) = Ax + b$, convergence to $x^* = -A^{-1}b$ is equivalent to A being Hurwitz (all eigenvalues have negative real parts). The Hurwitz condition is both necessary and sufficient: if A has any eigenvalue with positive real part, the iteration diverges.

6 Lock-In to Attractors

When the ODE has multiple locally stable equilibria, the SA iteration converges to one of them. The following result characterizes which equilibria are reachable.

Proposition 6.1 (Lock-In to Attractors)**.** *Suppose the ODE $\dot{x} = h(x)$ has finitely many equilibria, each either asymptotically stable or unstable. Under the ODE method assumptions (B1)–(B4) from Lesson 31, the SA iterates converge a.s. to one of the* stable *equilibria. The probability of converging to an unstable equilibrium is zero.*

Proof. By the ODE method, the limit set of $\{x_n\}$ is a compact connected ODE-flow-invariant set. By the avoidance of unstable equilibria result (Pemantle, 1990; Brandiere, 1998): with nondegenerate noise ($\mathbb{E}[M_{n+1}M_{n+1}^T \mid \mathcal{F}_n]$ positive definite), the probability of converging to an unstable equilibrium is zero. The noise "kicks" the iterate off the unstable manifold, and the ODE dynamics carry it toward a stable equilibrium. □

Multiple Stable Equilibria in RL

When the mean-field ODE has multiple stable equilibria, the RL algorithm may converge to a suboptimal one. This is a fundamental challenge in policy gradient methods where $J(\theta)$ is generally non-concave. Two-timescale SA (actor-critic) inherits this issue. Entropy regularization (as in SAC) and exploration bonuses are practical techniques to mitigate convergence to poor local optima.

7 RL Applications

We now map the general SA theory to specific RL algorithms.

7.1 TD(0) as Stochastic Approximation

TD(0) – Complete SA Analysis

Update. $V_{n+1}(s_n) = V_n(s_n) + \alpha_n(s_n)[R_{n+1} + \gamma V_n(s_{n+1}) - V_n(s_n)]$.
SA form. Linear SA with $A = \gamma P^\pi - I$, $b = r^\pi$. Since eigenvalues of P^π satisfy $|\lambda_i| \leq 1$ and $\gamma < 1$, all eigenvalues of A have real part at most $\gamma - 1 < 0$, so A is Hurwitz. By Theorem 5.4, TD(0) converges a.s. to $V^\pi = (I - \gamma P^\pi)^{-1} r^\pi$.
With linear FA: $V(s) = \phi(s)^T w$, $A = \Phi^T D(\gamma P^\pi - I)\Phi$ where $D = \mathrm{diag}(d^\pi)$. On-policy: A is negative definite, TD converges. Off-policy: A may not be Hurwitz — this is the deadly triad.

7.2 Q-Learning as Asynchronous SA

Q-Learning – Asynchronous SA

Update. $Q_{n+1}(s_n, a_n) = Q_n(s_n, a_n) + \alpha_n(s_n, a_n)[R_{n+1} + \gamma \max_{a'} Q_n(s_{n+1}, a') - Q_n(s_n, a_n)]$.
SA form. Nonlinear SA with drift $h(Q)(s, a) = r(s, a) + \gamma \sum_{s'} P(s'|s, a) \max_{a'} Q(s', a') - Q(s, a)$. The ODE $\dot{Q} = \mathcal{T}^* Q - Q$ uses the Bellman optimality operator $\mathcal{T}^*$, a γ-contraction. The Lyapunov function $V(Q) = \|Q - Q^*\|_\infty$ gives $\dot{V} \leq -(1 - \gamma)V$, so Q^* is globally stable. Under Robbins–Monro step sizes per (s, a) and sufficient exploration (each pair updated infinitely often), $Q_n \xrightarrow{\text{a.s.}} Q^*$.

7.3 REINFORCE as SA

REINFORCE – SA on the Gradient ODE

Update. $\theta_{n+1} = \theta_n + \alpha_n \widehat{\nabla J(\theta_n)}$.
SA form. This is Robbins–Monro with $h(\theta) = \nabla J(\theta)$ and noise $M_{n+1} = \widehat{\nabla J(\theta_n)} - \nabla J(\theta_n)$.
Associated ODE. $\dot{\theta} = \nabla J(\theta)$. Any local maximum θ^* with negative definite Hessian is a stable equilibrium. By Proposition 6.1, the iterates converge to a local maximum, avoiding saddle points and minima.
Convergence rate. The CLT (Theorem 3.3) gives $\sqrt{n}(\theta_n - \theta^*) \xrightarrow{d} \mathcal{N}(0, V)$ near each local maximum. The high variance of REINFORCE (Σ is large) motivates baselines: subtracting a baseline $b(s)$ from the return does not change $h(\theta)$ but reduces Σ, shrinking the asymptotic covariance V.

7.4 Actor-Critic as Two-Timescale SA

Actor-Critic – Two-Timescale Analysis

Critic w (fast, step b_n), actor θ (slow, step a_n), $a_n/b_n \to 0$:

$$w_{n+1} = w_n + b_n[\delta_n \phi(s_n)], \quad \delta_n = R_{n+1} + \gamma\phi(s_{n+1})^T w_n - \phi(s_n)^T w_n,$$
$$\theta_{n+1} = \theta_n + a_n[\delta_n \nabla_\theta \log \pi_\theta(a_n|s_n)].$$

Map: $y_n = w_n$ (fast), $x_n = \theta_n$ (slow). For fixed θ, the critic ODE is linear SA for TD under π_θ with equilibrium $\lambda(\theta) = w^*(\theta)$ ((E5)). The slow ODE becomes $\dot{\theta} \approx \nabla J(\theta)$ ((E6)). By Theorem 4.4, critic converges to $w^*(\theta^*)$ and actor converges to a local optimum θ^*.

8 Exercises

Exercise 8.1 (Polyak–Ruppert for Mean Estimation (*))**.** Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$. Consider the SA iteration $\theta_{n+1} = \theta_n + a_n(X_{n+1} - \theta_n)$ with $a_n = 1/n^{2/3}$.

(a) Show this is SA with $h(\theta) = \mu - \theta$ and identify A and Σ.

(b) Define $\bar{\theta}_n = \frac{1}{n}\sum_{k=1}^n \theta_k$. Using Theorem 3.5, find the asymptotic distribution of $\sqrt{n}(\bar{\theta}_n - \mu)$.

(c) Compare the asymptotic variance with that of the sample mean $\bar{X}_n$. Are they the same?

Exercise 8.2 (Two-Timescale Separation (**))**.** Consider the system

$$x_{n+1} = x_n + a_n(-x_n + y_n + M_{n+1}^{(1)}),$$
$$y_{n+1} = y_n + b_n(-2y_n + x_n + M_{n+1}^{(2)}),$$

with $a_n = 1/n$, $b_n = 1/n^{2/3}$, and noises i.i.d. with zero mean and unit variance.

(a) Verify $a_n/b_n \to 0$.

(b) Fix x and find $\lambda(x)$ such that $-2\lambda(x) + x = 0$.

(c) Substitute $y = \lambda(x)$ into the x-ODE to get $\dot{x} = -x/2$. Find x^*.

(d) Compute (x^*, y^*) and verify Hurwitz conditions for both fast and slow ODEs.

Exercise 8.3 (CLT for Scalar SA (**))**.** For the scalar SA $x_{n+1} = x_n + (a/n)(-\lambda x_n + M_{n+1})$ with $\lambda > 0$, $\mathbb{E}[M_{n+1}^2] = \sigma^2$:

(a) Verify that $x^* = 0$ and $A = -\lambda$.

(b) Find the condition on a for (D3) to hold.

(c) Solve the Lyapunov equation (1) for V.

(d) Find the optimal a minimizing V and compute V_{opt}.

(e) Compare with Polyak–Ruppert averaging applied to this problem.

Exercise 8.4 (Hurwitz Verification for TD (**))**.** Three-state MDP with transition matrix

$$P^\pi = \begin{pmatrix} 0.2 & 0.5 & 0.3 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.3 & 0.3 \end{pmatrix},$$

$\gamma = 0.95$, stationary distribution $d^\pi = (0.2, 0.45, 0.35)$, and feature matrix $\Phi = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$.

(a) Compute $A = \Phi^T D(\gamma P^\pi - I)\Phi$ where $D = \mathrm{diag}(d^\pi)$.

(b) Verify A is negative definite.

(c) Conclude TD(0) converges with this function approximation.

Exercise 8.5 (The Deadly Triad (***))**.** Construct a two-state MDP with off-policy sampling where TD(0) with linear FA diverges:

(a) Take $\mathcal{S} = \{1, 2\}$, $\gamma = 0.99$, behavior distribution $\mu(1) = 0.01$, $\mu(2) = 0.99$.

(b) Target policy transitions: $P^\pi(1|1) = 0.9$, $P^\pi(2|1) = 0.1$, $P^\pi(1|2) = 0.1$, $P^\pi(2|2) = 0.9$.

(c) Features: $\phi(1) = 1$, $\phi(2) = 2$ (scalar).

(d) Compute $A = \mathbb{E}_\mu[\phi(s)(\gamma\phi(s') - \phi(s))]$ and show it has a positive eigenvalue.

(e) Explain why the SA iteration diverges and relate to the Borkar–Meyn stability criterion.

Exercise 8.6 (GTD2 as Two-Timescale SA (***))**.** The GTD2 algorithm uses the updates:

$$\begin{aligned} w_{n+1} &= w_n + b_n[\phi_n(\phi_n - \gamma\phi'_n)^T\theta_n - \phi_n\phi_n^T w_n], \\ \theta_{n+1} &= \theta_n + a_n[\delta_n\phi_n - \gamma\phi'_n(\phi_n^T w_n)], \end{aligned}$$

where $\phi_n = \phi(s_n)$, $\phi'_n = \phi(s_{n+1})$, $\delta_n = R_{n+1} + \gamma\phi_n'^T\theta_n - \phi_n^T\theta_n$.

(a) Identify fast and slow iterates and verify timescale separation.

(b) Compute the fast-timescale equilibrium $\lambda(\theta)$.

(c) Substitute into the slow ODE and show it is gradient descent on the MSPBE (mean squared projected Bellman error).

(d) Explain why GTD2 avoids the deadly triad.

Exercise 8.7 (Linear SA with Non-Hurwitz Matrix (*))**.** Consider the scalar SA $x_{n+1} = x_n + a_n(\lambda x_n + M_{n+1})$ with $\lambda > 0$ and $a_n = 1/n$.

(a) Show that $A = \lambda$ is not Hurwitz and the ODE $\dot{x} = \lambda x$ has no stable equilibrium.

(b) Argue that $x_n \to \pm\infty$ a.s.

(c) Relate this to off-policy TD(0) where the SA matrix has a positive eigenvalue.

9 Summary

- **CLT for SA:** $\sqrt{n}(x_n - x^*) \xrightarrow{d} \mathcal{N}(0, V)$ with V from the Lyapunov equation. Rate: $O(1/\sqrt{n})$.
- **Polyak–Ruppert averaging:** $\bar{x}_n$ achieves optimal covariance $A^{-1}\Sigma A^{-T}$ without knowing A.
- **Two-timescale SA:** Fast iterate tracks $\lambda(x_n)$; slow iterate converges via $\dot{x} = f(x, \lambda(x))$.
- **Linear SA:** Convergence iff A Hurwitz. Boundedness automatic.
- **Lock-in:** SA avoids unstable equilibria.
- **RL:** TD(0) = linear SA (Hurwitz on-policy); Q-learning = nonlinear SA (contraction); REINFORCE = gradient ascent SA (lock-in); actor-critic = two-timescale SA.

10 Further Reading

- Lesson 31: Robbins–Monro, ODE method, Borkar–Meyn stability.
- Borkar [2008], Chapters 7–9: differential inclusions, constrained SA.
- Borkar [1997]: founding two-timescale SA paper.
- Polyak and Juditsky [1992]: original averaging proof with extensions.

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